

DHCP & VLANS Project Documentation

Items used for this project:

Copper straight through wires x10

ISR4331 Router x1

2960 IOS15 Switches x2

8 PCs x1

I am attempting to install a DHCP network that's segmented into different VLANs representing a department within a company.

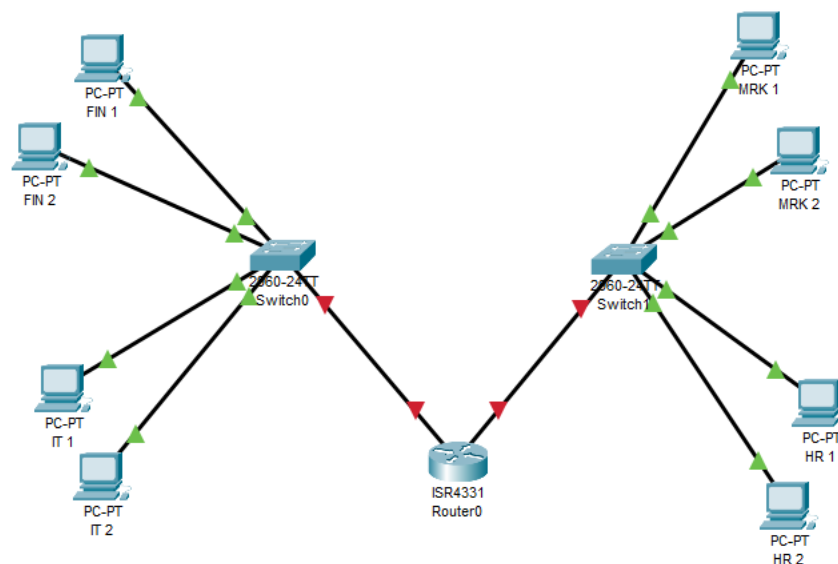


Figure 1: Beginning network map before configurations

In beginning the configurations for Switch0, I did the following commands highlighted in figures 2-4 to setup a VLAN for the finance department and a VLAN for the IT department.

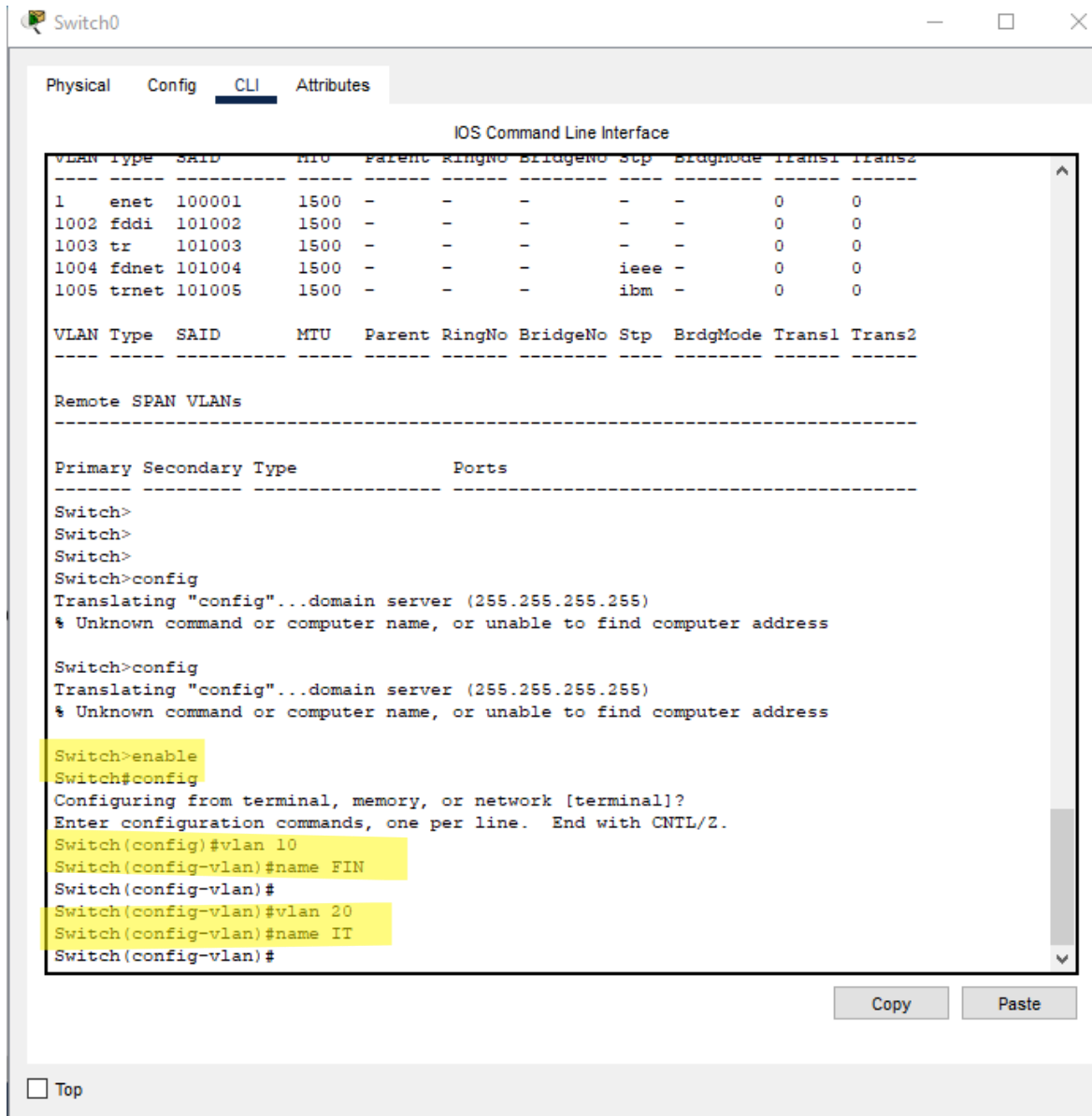


Figure 2: Setting up Switch0 Vlans

```

Switch#enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa0/1
Switch(config-if)#no shutdown
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#
Switch(config-if)#
Switch(config-if)#^Z
Switch#
%SYS-5-CONFIG_I: Configured from console by console
config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa0/2
Switch(config-if)#no shutdown
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10

```

Figure 3: Setting computer connected to switch0 Fa0/1 & Fa0/2 ports to vlan 10

```

Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa0/3
Switch(config-if)#no shutdown
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#^Z
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa0/4
Switch(config-if)#no shutdown
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#^Z
Switch#
%SYS-5-CONFIG_I: Configured from console by console

```

Figure 4: Setting computer connected to switch0 Fa0/3 & Fa0/4 ports to vlan 20

```
Switch#sh vlan
```

VLAN Name	Status	Ports
1 default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10 FIN	active	Fa0/1, Fa0/2
20 IT	active	Fa0/3, Fa0/4
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

VLAN Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Transl	Trans2
1	enet	100001	1500	-	-	-	-	0	0
10	enet	100010	1500	-	-	-	-	0	0
20	enet	100020	1500	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	0	0

--More--

Figure 5: show vlan result for Switch0:

For Switch1, I setup a VLAN for the marketing department and a VLAN for the Human Resources department as demonstrated below:

```
Switch>enable
Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa0/1
Switch(config-if)#no shutdown
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 11
Switch(config-if)#^Z
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa0/2
Switch(config-if)#no shutdown
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 11
Switch(config-if)#^Z
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa0/3
Switch(config-if)#no shutdown
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 21
Switch(config-if)#^Z
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa0/4
Switch(config-if)#no shutdown
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 21
Switch(config-if)#^Z
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
```

Figure 6: Setting up computers to their respective vlans in Switch1

Now, I went to the CLI of the router to properly set up the connection between both switches as shown below:

```
Router>enable
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int gi0/0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

Router(config-if)#^Z
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int gi0/0/1
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
```

Figure 7: Made router connection to the 2 switches up

I then had to encapsulate the connections of the VLANs from the router.

```
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int gi0/0/0.1
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0.1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.1, changed state to up

Router(config-subif)#ip address 10.0.0.1 255.255.255.0

% Configuring IP routing on a LAN subinterface is only allowed if that
subinterface is already configured as part of an IEEE 802.10, IEEE 802.1Q,
or ISL VLAN.

Router(config-subif)#encapsulation dot1Q ?
<1-4094> IEEE 802.1Q VLAN ID
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#^Z
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int gi0/0/0.2
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0.2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.2, changed state to up

Router(config-subif)#ip address 20.0.0.1 255.255.255.0

% Configuring IP routing on a LAN subinterface is only allowed if that
subinterface is already configured as part of an IEEE 802.10, IEEE 802.1Q,
or ISL VLAN.

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#
Router(config-subif)#
Router(config-subif)#^Z
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

Figure 8: Encapsulation on Router the VLANs of Switch0

```
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int gi0/0/1.1
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1.1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1.1, changed state to up

Router(config-subif)#ip address 11.0.0.1 255.255.255.0

% Configuring IP routing on a LAN subinterface is only allowed if that
subinterface is already configured as part of an IEEE 802.10, IEEE 802.1Q,
or ISL VLAN.

Router(config-subif)#encapsulation dot1Q 11
Router(config-subif)#
Router(config-subif)#
Router(config-subif)#^Z
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int gi0/0/1.2
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1.2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1.2, changed state to up

Router(config-subif)#ip address 21.0.0.1 255.255.255.0

% Configuring IP routing on a LAN subinterface is only allowed if that
subinterface is already configured as part of an IEEE 802.10, IEEE 802.1Q,
or ISL VLAN.

Router(config-subif)#encapsulation dot1Q 21
Router(config-subif)#^Z
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

Figure 9: Encapsulation on Router the VLANs of Switch1

Now the router's to be configured as a DHCP server

```
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp pool ?
WORD Pool name
Router(config)#ip dhcp pool dhcpvlan10
Router(dhcp-config)#network 10.0.0.0 255.255.255.0
Router(dhcp-config)#de
% Incomplete command.
Router(dhcp-config)#default-router 10.0.0.1
Router(dhcp-config)#
Router(dhcp-config)#ip dhcp pool dhcpvlan20
Router(dhcp-config)#default-router 20.0.0.1
Router(dhcp-config)#
Router(dhcp-config)#ip dhcp pool dhcpvlan11
Router(dhcp-config)#default-router 11.0.0.1
Router(dhcp-config)#ip dhcp pool dhcpvlan21
Router(dhcp-config)#default-router 21.0.0.1
Router(dhcp-config)#
```

Figure 10: Router configurations for DHCP setup

```
Router(config)#ip dhcp pool dhcpvlan20
Router(dhcp-config)#network 20.0.0.0 255.255.255.0
Router(dhcp-config)#^Z
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp pool dhcpvlan11
Router(dhcp-config)#network 11.0.0.0 255.255.255.0
Router(dhcp-config)#^Z
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#network 21.0.0.0 255.255.255.0
^
% Invalid input detected at '^' marker.

Router(config)#ip dhcp pool dhcpvlan21
Router(dhcp-config)#network 21.0.0.0 255.255.255.0
Router(dhcp-config)#^Z
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 11: Screenshot subnets I forgot to add for the DHCP router configuration for VLANs 20, 11, & 21

```

Router#sh ip dhcp pool

Pool dhcpvlan10 :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 0
Excluded addresses : 0
Pending event : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
10.0.0.1          10.0.0.1 - 10.0.0.254  0 / 0 / 254

Pool dhcpvlan20 :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 0
Excluded addresses : 0
Pending event : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
20.0.0.1          20.0.0.1 - 20.0.0.254  0 / 0 / 254

Pool dhcpvlan11 :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 0
Excluded addresses : 0
Pending event : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
11.0.0.1          11.0.0.1 - 11.0.0.254  0 / 0 / 254

Pool dhcpvlan21 :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 0
Excluded addresses : 0
Pending event : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
21.0.0.1          21.0.0.1 - 21.0.0.254  0 / 0 / 254
Router#

```

Figure 12: shown DHCP pool

Normally, the process of finishing up this DHCP network of VLANs would be finished by now, however there were some mistakes I made along the way that had me go back and troubleshoot as PCs were failing at getting a DHCP IP and defaulting to an APIPA address.

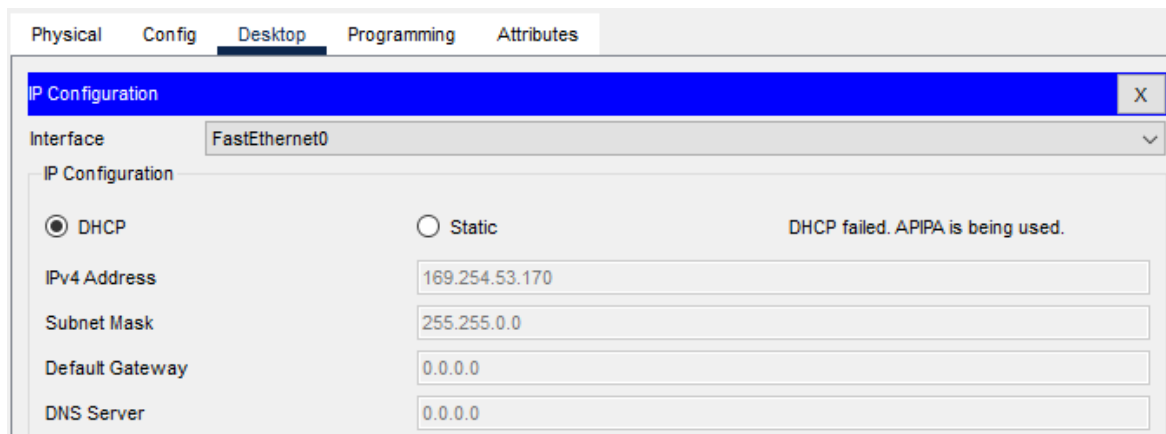


Figure 13: As of right now, DHCP is failing and a PC is getting APIPA instead, the following steps is how we will fix this issue

Using command **sh run | i dhcp** (short for **show running-config | include dhcp**) helped me find specific lines in the configuration that contain the word **"dhcp"**. I found that there was a mistake in not declaring ip addresses for interfaces to VLANs.

```
Router#sh run | i dhcp
ip dhcp pool dhcpvlan10
ip dhcp pool dhcpvlan20
ip dhcp pool dhcpvlan11
ip dhcp pool dhcpvlan21
Router#sh run | section dhcp
ip dhcp pool dhcpvlan10
network 10.0.0.0 255.255.255.0
default-router 10.0.0.1
ip dhcp pool dhcpvlan20
network 20.0.0.0 255.255.255.0
default-router 20.0.0.1
ip dhcp pool dhcpvlan11
network 11.0.0.0 255.255.255.0
default-router 11.0.0.1
ip dhcp pool dhcpvlan21
network 21.0.0.0 255.255.255.0
default-router 21.0.0.1
Router#sh run

interface GigabitEthernet0/0/0
no ip address
duplex auto
speed auto
!
interface GigabitEthernet0/0/0.1
encapsulation dot1Q 10
no ip address
!
interface GigabitEthernet0/0/0.2
encapsulation dot1Q 20
no ip address
!
interface GigabitEthernet0/0/1
no ip address
duplex auto
speed auto
!
interface GigabitEthernet0/0/1.1
encapsulation dot1Q 11
no ip address
!
interface GigabitEthernet0/0/1.2
encapsulation dot1Q 21
no ip address
!
interface GigabitEthernet0/0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Vlan1
Router#
```

Figure 14: Information show from the **sh run | ihcp** command

I quickly added the missing ip address ranges

```
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int gi0/0/0.1
Router(config-subif)#ip ad
% Incomplete command.
Router(config-subif)#ip address 10.0.0.1 255.255.255.0
Router(config-subif)#^Z
Router#

Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int gi0/0/0.2
Router(config-subif)#ip address 20.0.0.1 255.255.255.0
Router(config-subif)#int gi0/0/1.1
Router(config-subif)#ip address 11.0.0.1 255.255.255.0
Router(config-subif)#int gi0/0/1.2
Router(config-subif)#ip address 21.0.0.1 255.255.255.0
Router(config-subif)#
```

Figure 15: adding the IP address ranges for the VLANs

I still had issues after adding the missing ranges, so I decided to do further troubleshooting and after another quick diagnosis, I found that I forgot to trunk the switch ports that lead to the router

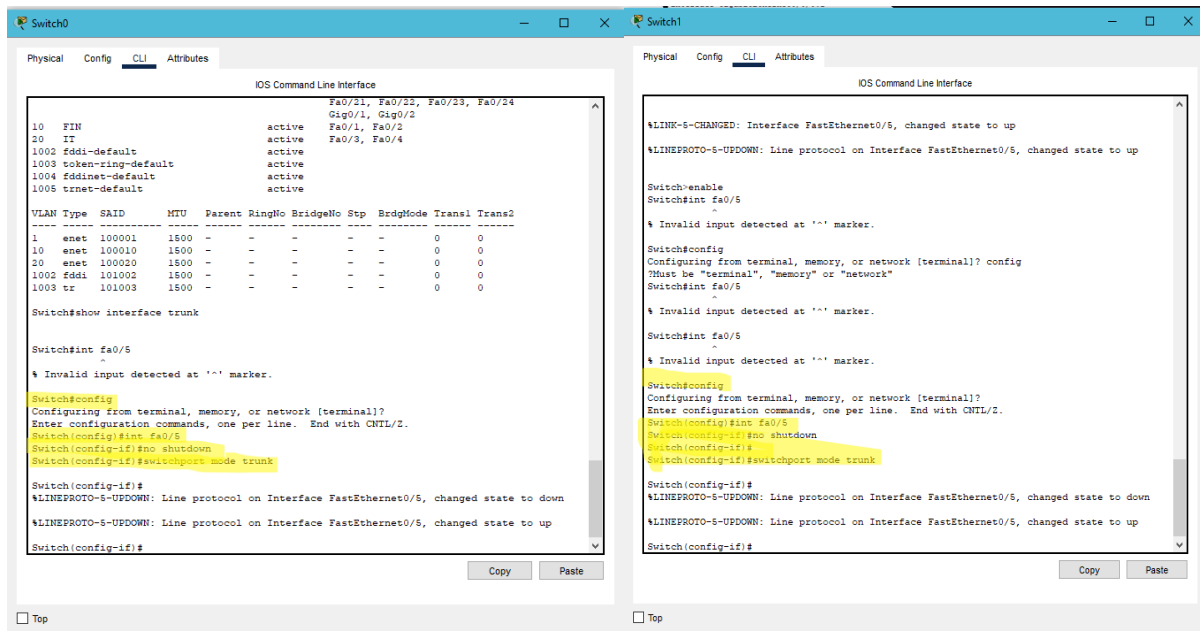


Figure 16: setting up the trunk configurations between the physical link of the switches to the router

In the end, I was able to get DHCP to be working properly and assigning the right IP addresses to each department.

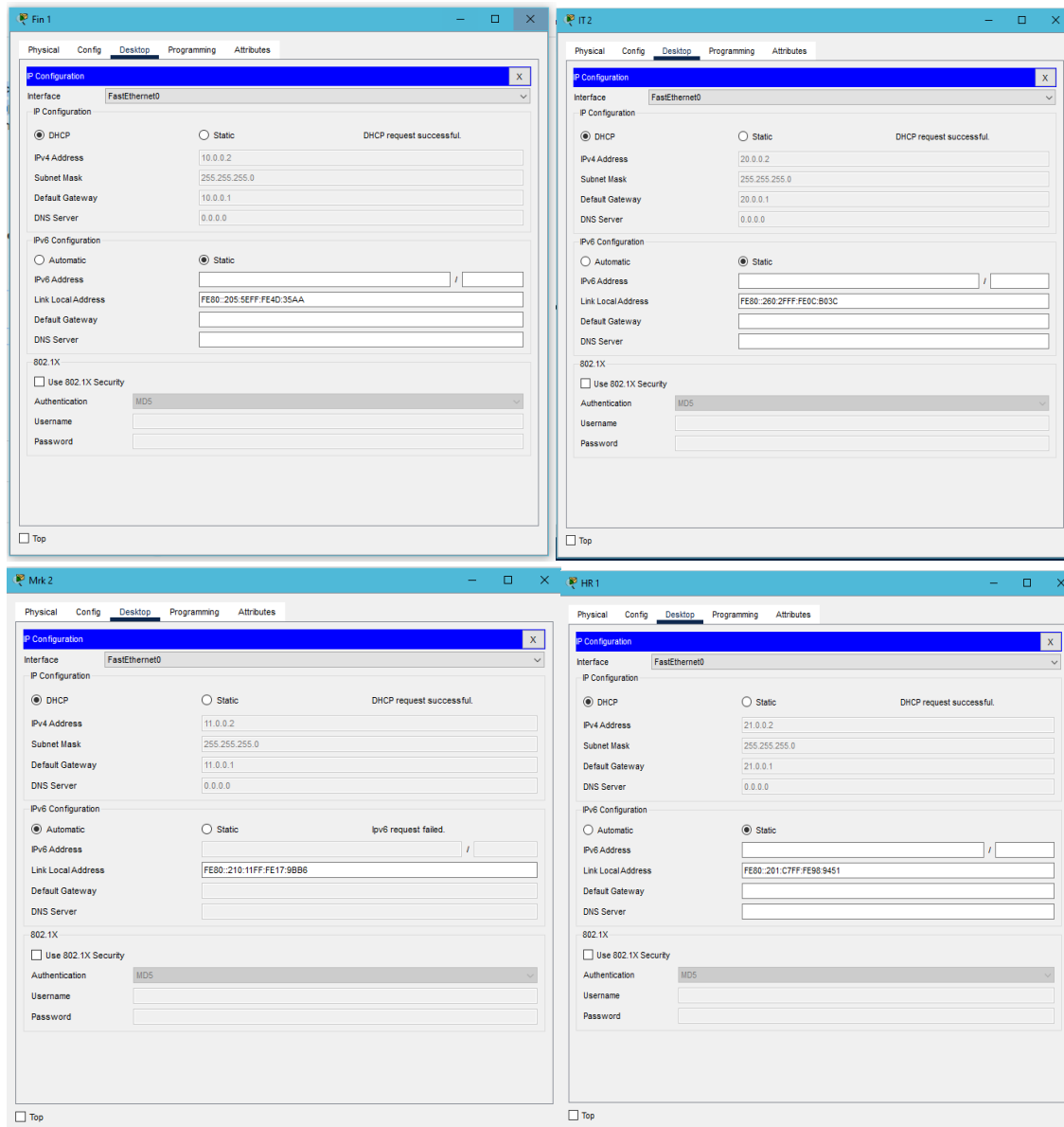


Figure 17: DHCP assignments from a computer from each department

Here is the look of my network diagram at the end of this project:

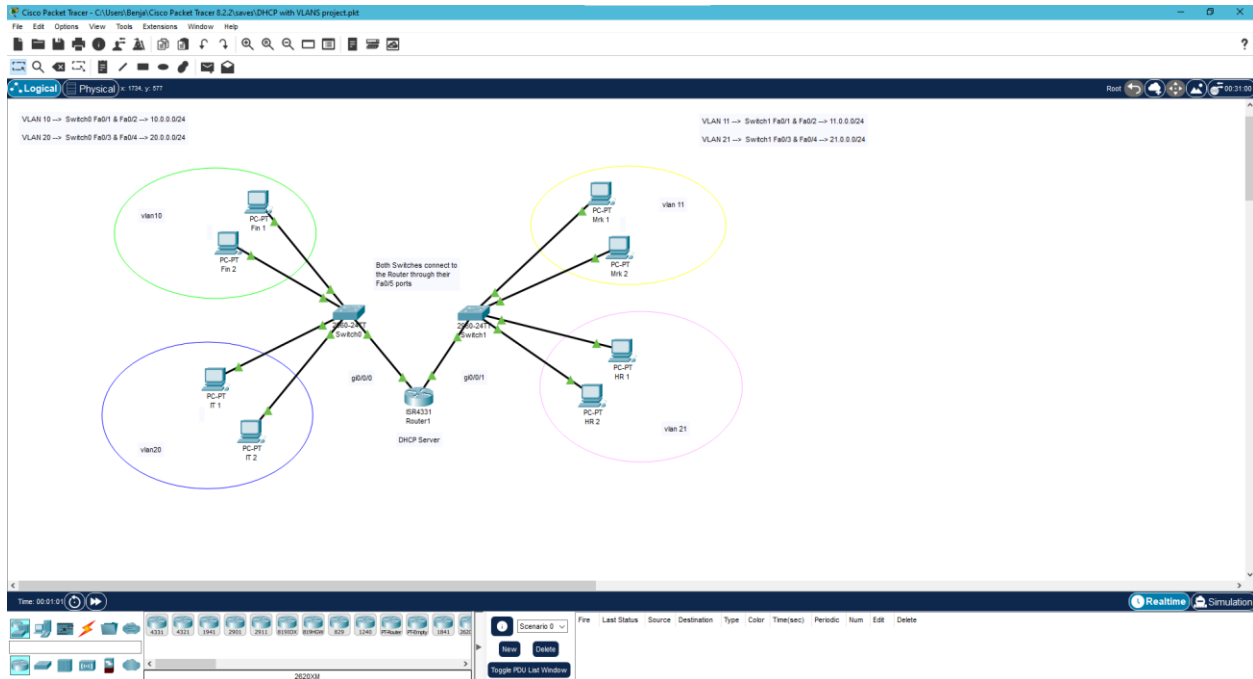


Figure 18: Final network Diagram showing working traffic flow